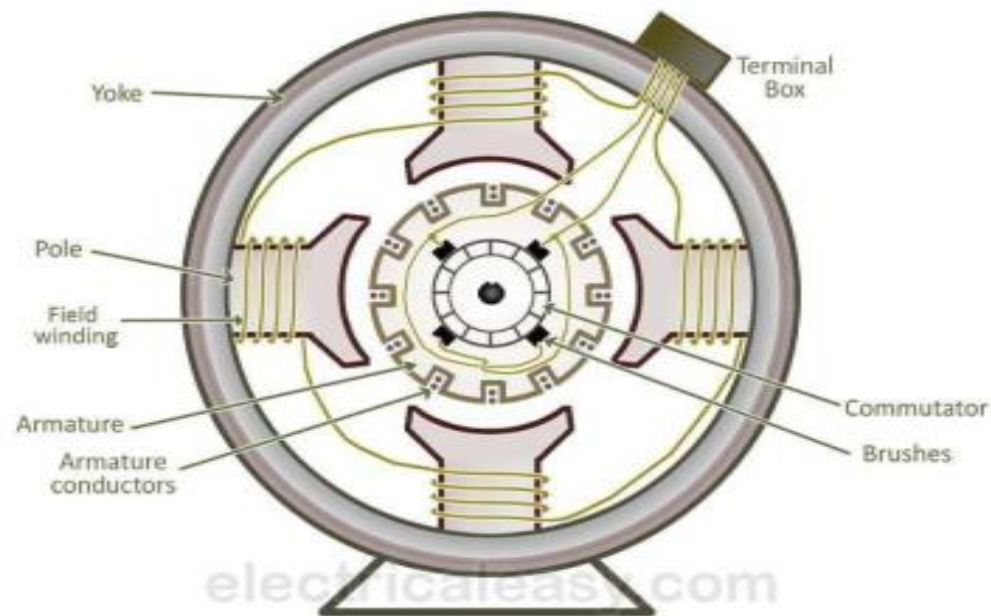


Syllabus : Construction, principle of operation, and applications of DC Machine (Motor & Generator), 1- ϕ Transformer, BLDC Motor, Stepper Motor, 1- ϕ Induction Motor.

DC Machine

A Machine which operates on DC Power supply is called DC Machine. DC Machine can be operated as Generator or Motor . If mechanical energy converted into electrical energy is called generator .If electrical energy converted into mechanical energy is called motor

DC Machine –Constructional Details



A DC machine consists of two basic parts, stator and rotor.

The other important parts are described below.

Yoke:

The outer frame of a D.C Machine (Generator or Motor) is called as yoke. Yoke is made up of cast iron or steel. Yoke provides mechanical strength for whole assembly of the D.C Machine It also carries the magnetic flux produced by the poles acting as magnetic path.

Poles:

Poles are to support field windings or coils which are wound around it. Poles are joined to the yoke with the help of screws or welding

Pole shoe:

Pole shoe is an extended part of the pole which serves two purposes, (i) To prevent field coils from slipping and (ii) To spread out the flux in air gap uniformly.

Field winding:

Field winding is wound on poles and connected in series or parallel with armature winding. Field coils are mounted on the poles and carry the dc exciting current due to which pole core on which the winding is placed behaves as an electromagnet, producing necessary flux. The field coils are connected in such a way that adjacent poles have opposite polarity (alternate N and S poles)

Armature core and Armature winding:

Armature core is the rotor of a D.C Machine. Armature core is cylindrical in shape on which slots are provided to carry armature winding. The armature core is laminated to reduce the eddy current loss. Armature winding can be wound by one of the two methods known as Lap winding ($A=P$) and Wave winding ($A=2$)

Commutator:

In DC Generator, commutator is a mechanical rectifier which converts the alternating voltage generated in the armature winding into direct voltage across the brushes. In DC Motor, commutator acts as mechanical inverter which converts direct voltage into

alternating voltage. The commutator is made up of copper segments insulated from each other by mica sheets and mounted on the shaft of the machine

Brushes:

The purpose of brushes is to ensure electrical connections between the rotating commutator and stationary external load circuit. The brushes are made of carbon or graphite and rest on the commutator. Thus brushes are physically in contact with armature conductors hence wires can be connected to brushes

Principle of DC Generator

It is an electrical machine which converts mechanical energy into electrical energy.

The DC Generator works on the principle of Faraday's law of electromagnetic induction. The principle of operation of the dc generator is based on dynamically induced emf.

Whenever a rotating conductor cuts magnetic flux, an emf will be induced in the conductor.

The direction of induced emf in the conductor can be found by applying Fleming right hand rule

Fleming's right-hand rule

According to Fleming's right-hand rule, if the forefinger, middle finger, and thumb of the right hand are stretched naturally perpendicular to each other as shown in Fig 1.1, the forefinger indicates the direction of the magnetic field, the middle finger indicates the direction of the induced emf, and the thumb indicates the direction of motion of the conductor.

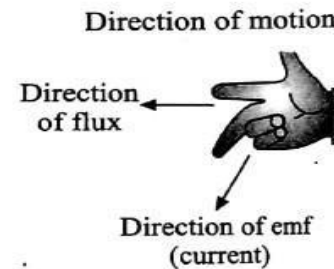


Fig1.1 Fleming's right-hand rule

DC Motor Principle of Operation:

Whenever a current carrying conductor is placed in the magnetic field it experiences a force. The direction of force can be found by applying Fleming's left-hand rule.

Fleming's left-hand rule

According to Fleming's left-hand rule, if the forefinger, middle finger, and thumb of the left hand are stretched naturally perpendicular to each other as shown in Fig 1.2, the forefinger indicates the direction of the magnetic field, the middle finger indicates the direction of the current, and the thumb indicates the direction of the force developed.

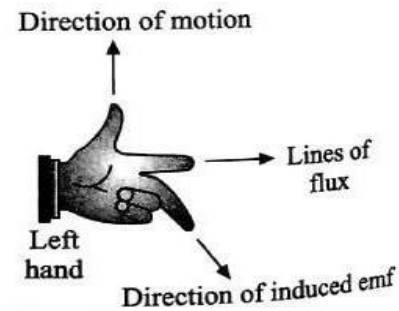
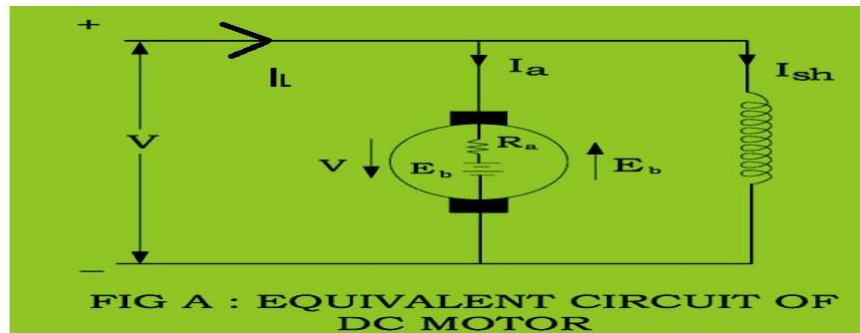


Fig 1.2 Fleming's left-hand rule

Back EMF

According to fundamental laws of nature, no energy conversion is possible until there is something to oppose the conversion. In case of generators this opposition is provided by magnetic drag, but in case of dc motors there is back emf. The direction of induced emf is found by Fleming's Right hand rule which is in direct opposition to the supply voltage. This emf is referred to as back emf or counter emf.

The expression of the back emf is given by $\text{Back emf } (E_b) = \Phi ZN / 60 \times (P / A) \text{ Volts}$



Significance of Back EMF

1. Magnitude of back emf is directly proportional to speed of the motor.
2. Consider the load on a dc motor is suddenly reduced. In this case, required torque will be small as compared to the current torque. Speed of the motor will start increasing due to the excess torque. Hence, being proportional to the speed, magnitude of the back emf will also increase.
3. With increasing back emf armature current will start decreasing
4. Torque being proportional to the armature current, it will also decrease until it becomes sufficient for the load. Thus, speed of the motor will regulate.
5. On the other hand, if a dc motor is suddenly loaded, the load will cause decrease in the speed. Due to decrease in speed, back emf will also decrease allowing more armature current.
6. Increased armature current will increase the torque to satisfy the load requirement. Hence, presence of the back emf makes a dc motor 'self- regulating

DC Generators applications

Type of DC Generator	Main Applications
Separately excited	:Exciters for alternators, lab experiments
Shunt wound	; Battery charging, electroplating, excitation of alternators
Series wound	: Boosters, arc welding
Cumulative compound	: General power supply, lighting, heavy loads
Differential compound	: Rare, sometimes arc welding

DC Motor Applications

<i>Type of motor</i>	<i>Characteristics</i>	<i>Applications</i>
Shunt	Approximately constant speed Adjustable speed Medium starting torque (Up to 1.5 F.L. torque)	For driving constant speed line shafting Lathes Centrifugal pumps Machine tools Blowers and fans Reciprocating pumps
Series	Variable speed Adjustable varying speed High Starting torque	For traction work <i>i.e.</i> , Electric locomotives Rapid transit systems Trolley, cars etc. Cranes and hoists Conveyors
Comulative Compound	Variable speed Adjustable varying speed High starting torque	For intermittent high torque loads For shears and punches Elevators Conveyors Heavy planers Heavy planers Rolling mills; Ice machines; Printing presses; Air compressors

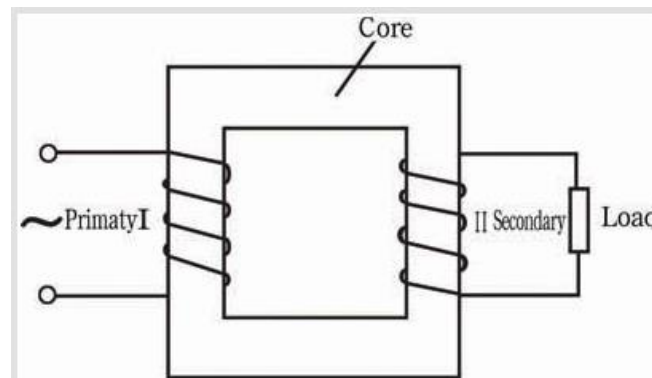
Single phase Transformer

An electrical power transformer is a static device, which transforms electrical power from one circuit to another circuit and transforms voltage one level to another level without changing frequency.

Constructional Details; the transformer has mainly two parts (1) core (II) Winding

Core: It is made-up of material called silicon steel material with laminations. To have high permeability the core is designed with Cold-rolled, grain-oriented (CRGO) silicon steel sheets (3–5% Si content). The core must have High permeability, Low hysteresis loss and Low eddy current loss .The Purpose of Transformer Core is

- The core provides a low-reluctance magnetic path for the magnetic flux linking the primary and secondary windings.
- It ensures efficient transfer of energy from one winding to another by electromagnetic induction.
- Without a core, the flux would pass mostly through air (high reluctance), causing poor efficiency.



Transformer Winding: Winding of transformer is made up of copper material. Transformer has two windings Primary winding and secondary winding. The winding which is connected to input ac supply is called primary winding. The winding which is connected to output or load is called secondary winding. Based on the input and output voltages the number of turns on the windings changes. The numbers of turns in primary winding is denoted by N_1 and Numbers of turns in secondary winding is denoted by N_2 . The winding which has more number of turns is called high voltage winding and the winding which has less number of turns is called Low voltage winding.

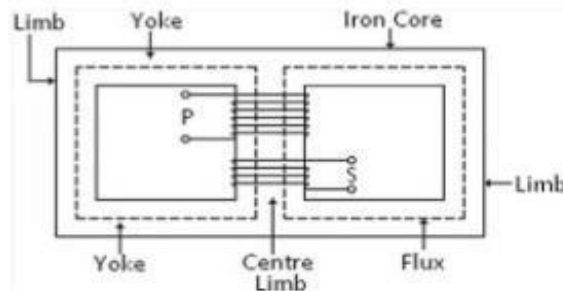
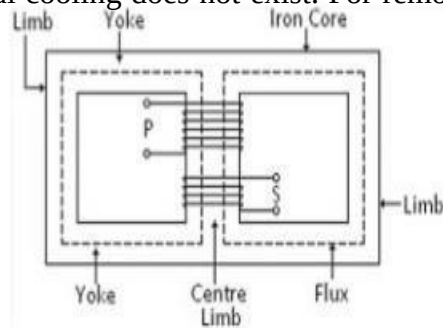
Types of transformers

On the basis of construction, the transformers are classified in 2 types :

- 1) Shell Type Transformer,
- 2) Core Type Transformer

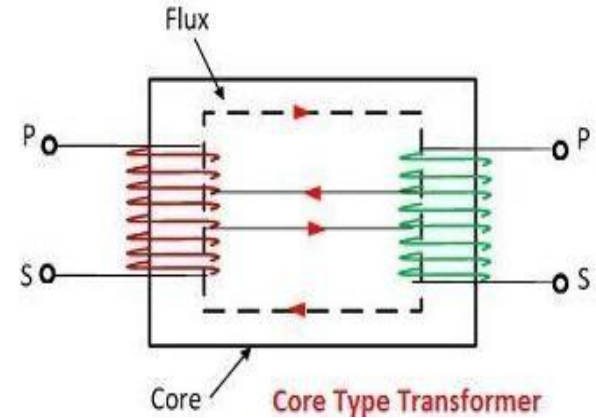
1. Shell Type Transformer :

For the construction of Shell type transformer a three limbed iron is used. Both primary and secondary windings are placed on central limb. First primary winding is wound on the central limb with required number of turns. Then insulating paper is wrapped on this primary winding, and then secondary winding is wound over this paper. It has double magnetic circuit. The core encircles most of the part of winding. Generally it is preferred for low voltage transformers. As the windings are surrounded by core, the natural cooling does not exist. For removing any winding for maintenance, large number of laminations is required to be removed

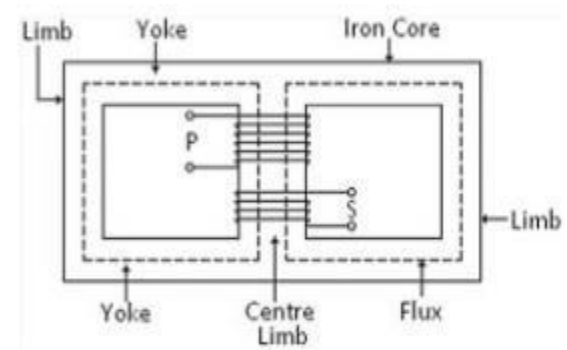
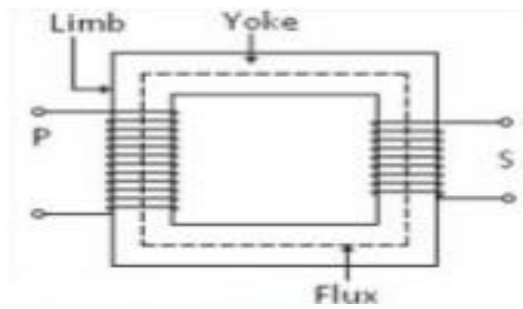


2) Core Type Transformer

For core type of transformer two limbed iron core is used. Primary and secondary winding turns are split in two parts and each part is arranged on both the limbs of core. Initially first part of the primary windings is wound on other limb, then insulating paper is wrapped on each part of primary winding. Now secondary winding is wound on this paper in two parts, each part on one limb. It has a single magnetic circuit. The winding encircles the core i.e. Core is surrounded by the winding. Core is made up of large number of thin laminations. As the windings are uniformly distributed on the two limbs the natural cooling is more effective. The coils can be easily removed by removing the laminations of the top yoke for maintenance.



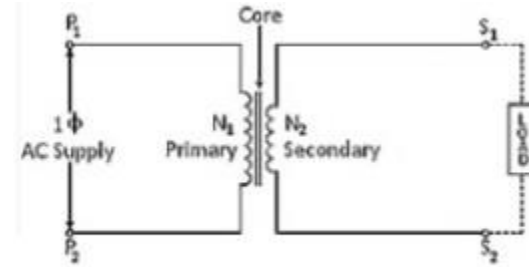
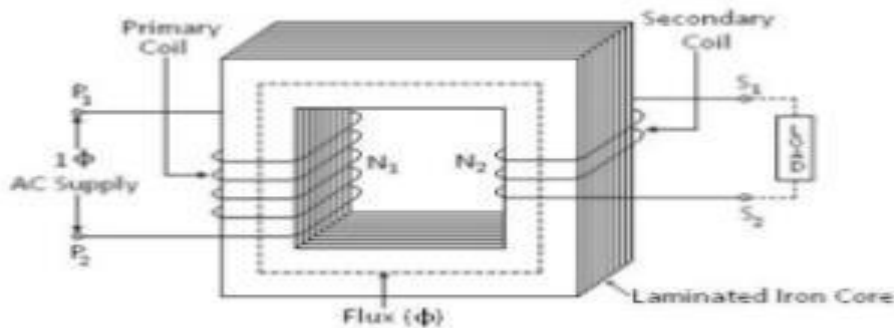
Comparison of Core and Shell Type



S. No	Core Type	Shell Type
1	The winding encircles the core	The core encircles most part of the winding
2	It has single magnetic circuit.	It has a double magnetic circuit .
3	The core has two limbs.	The core has three limbs.
4	The cylindrical coils are used.	The multilayer disc or sandwich type coils are used.
5	The windings are uniformly distributed on two limbs hence natural cooling is effective.	The natural cooling does not exist as the windings are surrounded by the core.
6	The coils can be easily removed from maintenance point of view.	The coils can not be removed easily
7	Preferred for high voltage transformers.	Preferred for low voltage transformers.

Working Principle of Transformer

1) A transformer operates on the principle of mutual inductance between two inductively coupled coils. The winding in which electrical energy is fed is called Primary winding and the other from which energy is drawn out is called Secondary winding.



2) The primary winding has 'N1' number of turns, while the secondary winding has N2 number of turns. The basic transformer and its symbol is shown in fig.(1) and fig (2) respectively.

3) When an alternating voltage V1 is applied to primary winding, an alternating current I1 flows in it producing alternating flux in the core. According to Faraday's law of electromagnetic induction an e.m.f. is induced in the primary winding which is given by,

$$E = -N \frac{d\Phi}{dt}$$

where N1 is the number of turns in primary winding. The induced e.m.f in the primary winding is nearly equal and opposite to the applied voltage V1.

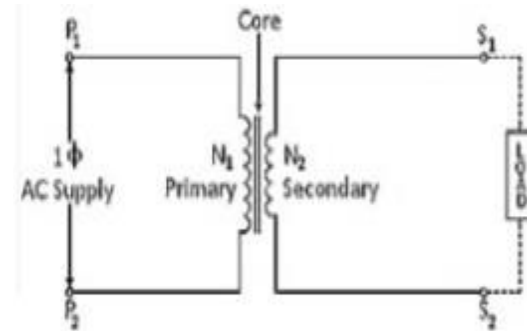
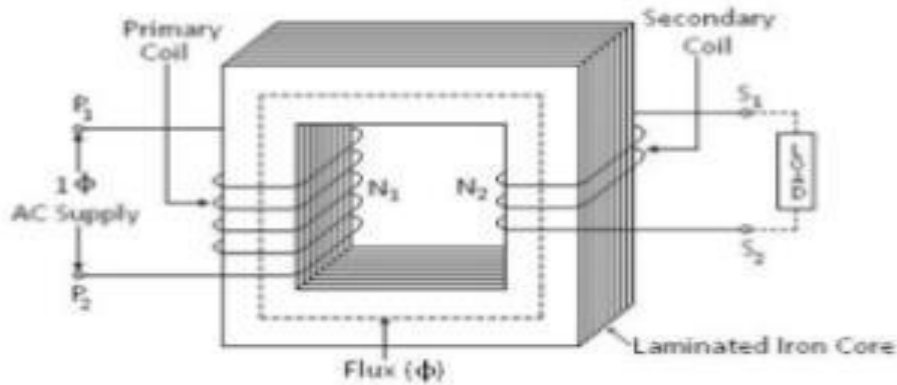


Fig.(2)

4) Assuming leakage flux to be negligible, almost whole flux produced in primary winding links with the secondary winding. Hence e.m.f. e_2 is induced in the secondary winding.

According to their working

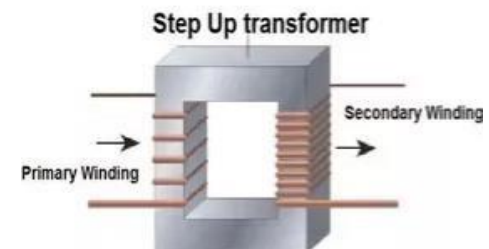
On the basis of working, the transformers are classified in three types:

1) Step up Transformer 2) Step down Transformer 3) Isolation Transformer

1) Step up Transformer: When the number of turns of secondary winding (N_2) are greater than number of turns of primary windings (N_1), then voltage available across secondary is greater than the applied across primary.

This kind of transformer is known as Step up Transformer

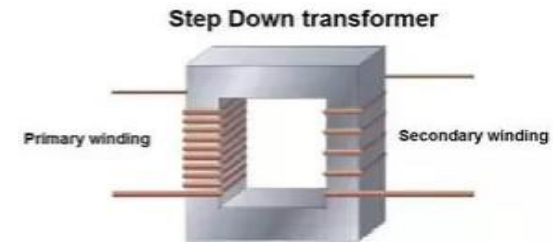
($K > 1$) where $K = N_2/N_1$ These type of transformers are generally used in generating power stations or in communication circuits as EHT Transformers



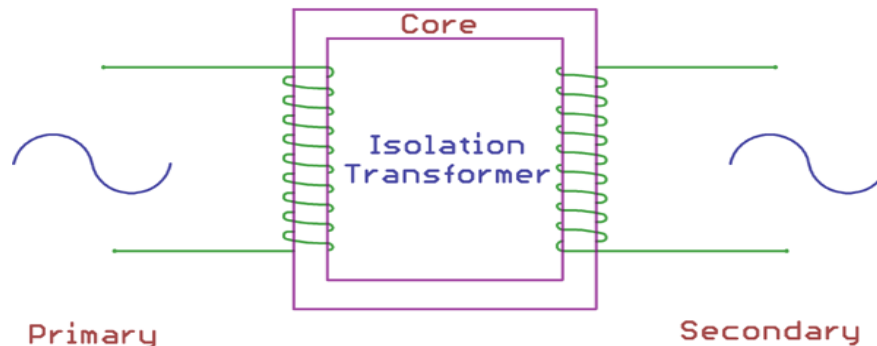
2) Step down Transformer : When the number of turns of secondary winding (N_2) are less than number of turns of primary windings (N_1), then voltage available across secondary is less than the applied across primary.

This kind of transformer is known as Step down Transformer

($K < 1$) where $K = N_2/N_1$. These type of transformers are generally used in distribution power stations and in electronics and communication circuits.



3. Isolation Transformer: 1) When the number of turns of secondary winding (N_2) is equal to the number of turns of primary windings (N_1), then voltage available across secondary is equal to voltage applied across primary.



This kind of transformer is known as isolation transformer ($K = 1$) where $K = N_2/N_1$. These type of transformers are mainly used to isolate the electrical or electronics circuits under testing from supply

BLDC Motor

The most severe problems associated with PMDC motor is the commutation with the brushes BLDC motors eliminate this problems as they have to commutator and brushes.

In BLDC motors the armature is stationary and permanent magnet field system is mounted on the rotating shaft. The synchronous motor with permanent magnet rotor and is supplied current from dc source through an inverter.

BLDC drive is comparable with induction motor drive up to 20kw.compared with PMDC motors, BLDC motors have higher efficiency, smaller size and better cooling.

Classification of BLDC Motors

4.1 On the basis of rotor construction BLDC motors

(i) Surface mounted permanent magnet (ii) Interior type permanent magnet motors.

Construction:

Stator

The stator of the BLDC motor is similar to that of a conventional induction motor .the stator core is made up of silicon steel stampings which are stacked together and fixed on the stator frame.

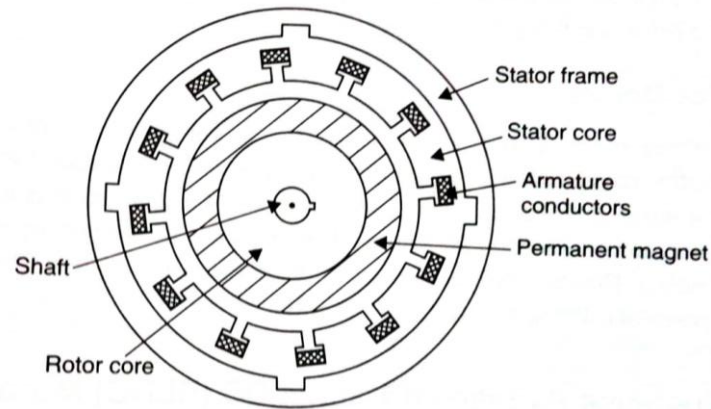
The stator core has slots in the inner periphery to house the armature conductors. closed or open type winding is used.

The armature winding is connected to the dc source through suitable semiconductor switching circuit.

Rotor

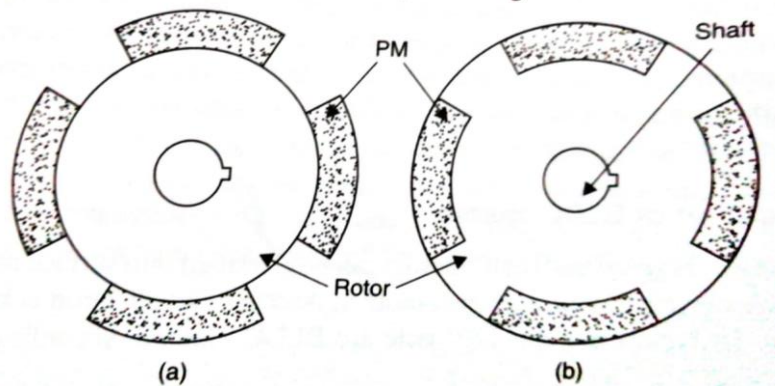
The rotor is made up of forged steel and carries permanent magnets .the armature should be wound for same number of permanent magnets. The rotor core is mounted on the shaft for successful commutation, the rotor position should be feedback.

For this purpose, a position sensor is to be mounted on the shaft. The sensor output is used to generate appropriate signals for triggering power electronic switches in the inverter circuit that feeds the BLDC motor



The permanent magnet can be fixed on the rotor in two different ways, resulting in surface mounted permanent magnet rotor.

If the magnets are mounted on the surface of the rotor, it is known as surface mounted permanent magnet rotor. In interior pm rotor, the magnets are placed in groves or slots as shown in fig. b.



Compared with the surface mounted pm motor, the interior pm rotor has the advantages like robust construction ability to operate at high speeds, smaller effective air gap length and higher torque to ampere ratio

3 Principle of operation

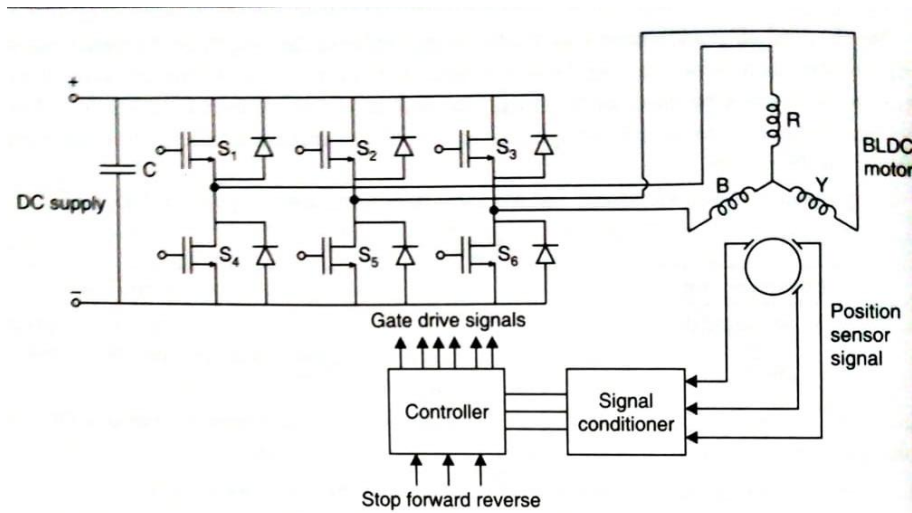
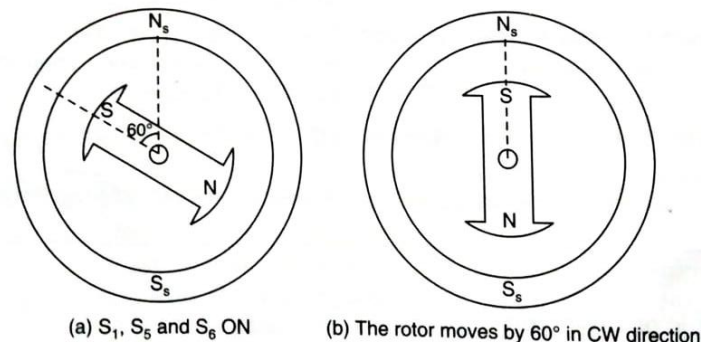


Fig. 3.16 Drive circuit for BLDC motor



Consider a BLDC motor supplied by voltage fed inverter as shown in fig 3.16. The inverter converts the dc voltage into variable frequency voltage. With this circuit, four quadrant operations are possible.

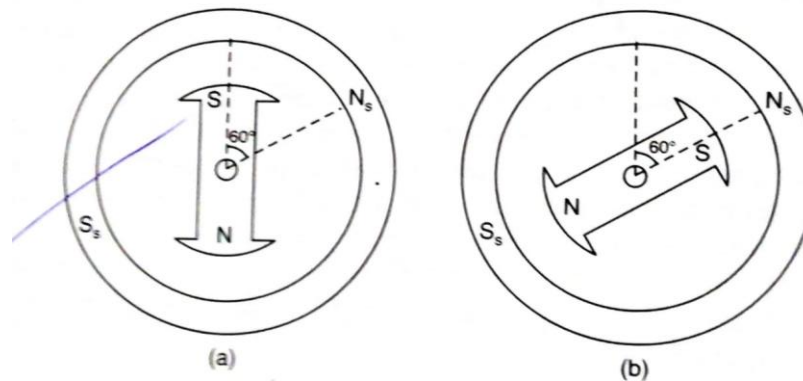
The harmonics generated by the inverter is blocked by the filter capacitor by the filter capacitor. The motor is provided with position sensor .the sensor provides necessary signals for switching the appropriate power switches of the inverter.

The switching is done in such a way that all the three phase's conduct at all time. This is required for obtaining 180° conduction. The switching interval is 60° . Due to the spatial arrangement of the armature winding, the mmf; produced by the stator rotates in space.

Assume that initially the rotor magnetic poles are in the position shown in fig 3.16.with this position of rotor, switches S1, S5 and S6 are turned on. the current flows through phase R and divides into equal amounts and completes its path through phase Y, phase B,S6 and S5.this current produces MMFS which can be the represented by stator poles experience torque and move in clockwise direction.

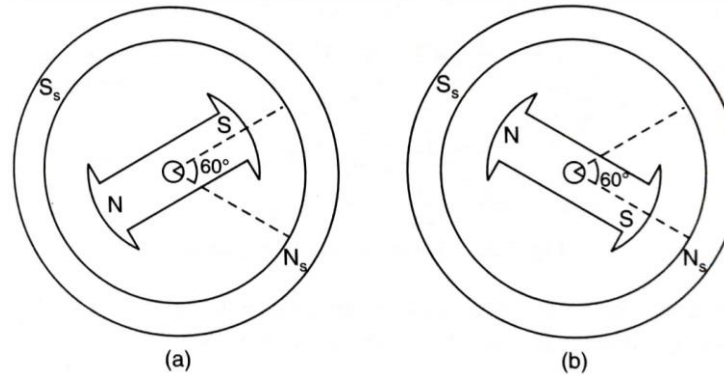
When the rotor moves by 60° and occupies the position shown in fig. switches S2 is turned on and S5 is turned off keeping S1 and S6 ON.

When S1, S2 and s6 are conducting with R and Y connected to positive and B to negative of power supply, the stator poles are shifted fare shifted by 60 as shown in fig 3.18a.this will cause the movement of rotor by 60° in clockwise direction and its position as shown in fig 2



Rotor movement when S1, S2, and S6 ON

When the rotor attains the position shown in fig3.18b, S1 is switched off and S4 is switched on. This operation results the position shown in fig. This will produce torque on the rotor and hence, it rotates by 60° when the next switching operation is done. This is shown in fig 3



Thus for the forward(clockwise) rotation of the BLDC motor, the switches should be operated in the sequence 1, 2, 6; 2, 4, 6; 2, 3, 4; 3, 4, 5; 1, 3, 5; 1, 5, 6; 1, 2, 6;....for reversing the direction the sequence is 1, 2, 6; 1, 5, 6; 1, 3, 5; 3, 4, 5; 2, 3, 4; 2, 4, 6; 1, 2, 6, for stopping ,keep the switches in the same condition (on/off) corresponding to the last operation.

Thus, when supply is switched on, current flows through the armature winding whose distribution depends on rotor position and the switching ON of the devices at various instants. Due to the interaction between current and magnetic field, rotor experiences torque and if this torque is greater than the load torque, rotor starts moving. BLDC motor is thus a self-starting motor

STEPPER MOTORS

1. Introduction

A stepper motor is an electro mechanical device which rotates in discrete angular steps in response to the input electrical pulses.

The power range of stepper motor is 1W to 2.5KW and torque range is about 1 micro N-m to 40 N-m.

Principle

A stepper motor works on the principle that a magnetic interaction between the rotor and stator which makes the stepping.

Types of stepper motors

Variable reluctance stepper motor (VR Stepper motor) - Single stack or multi-stack stepper motor

Hybrid Stepper motor Permanent magnet stepper motor

Applications:

- Printers
- Graph plotters
- Tape driver
- Disk Drives
- Machine Tools
- X-Y Recorders
- Robotics space Vehicle
- IC Fabrication and Electric Watches

Variable Reluctance (VR) STEPPER MOTORS

It has a stator with salient poles that have concentric windings which form different phases. The stator phases are excited by current pulses to establish a magnetic field.

The rotor has salient structure with projecting teeth and has no winding on permanent magnet.

Single Stack VR Stepper Motors

Construction

- The stator of a single stack VR stepper motors is made up of silicon steel stampings. The poles carry concentric windings. In stator with even number of poles, the windings on opposite poles are connected in series such that the currents in them establish magnetic fluxes that aid each other.
- Rotor is made of laminated silicon steel. It has no windings or permanent magnets. It has six rotor teeth.

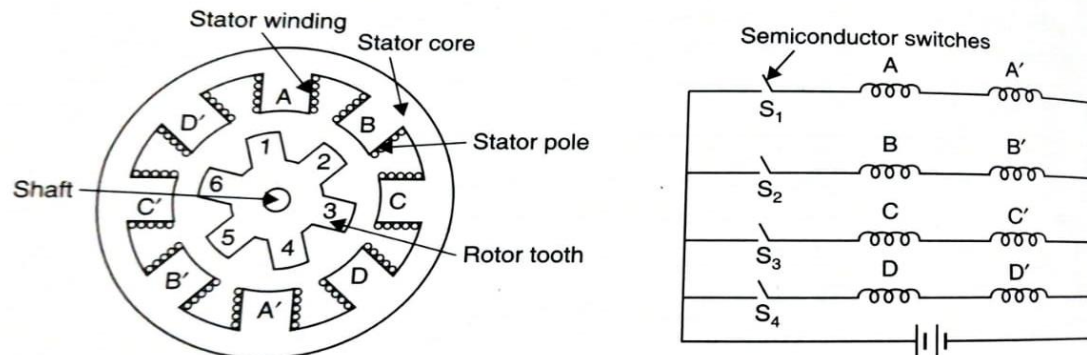


Fig. 1 Four-phase, eight-pole single-stack VR stepper motor and switching circuit

Working:

VR Stepper motor works on the principle that a magnetic material placed in a magnetic field experiences a force align it in a path of minimum reluctance.

Step angle $\theta_s = \frac{360}{mN_r}$

Where m = number of stator phases, N_r = number of rotor teeth.

In this case, $m = 4, N_r = 6$

$$\theta_s = \frac{360^\circ}{4 \times 6} = 15^\circ$$

Truth table

S ₁	S ₂	S ₃	S ₄	Θ
1	0	0	0	0°
0	1	0	0	15°
0	0	1	0	30°
0	0	0	1	45°
1	0	0	0	60°

1 – ON; 0 – OFF

Stator pole pitch: $\frac{360^\circ}{\text{No of stator pole teeth}} = \frac{360^\circ}{8} = 45^\circ$

Rotor pole pitch: $\frac{360^\circ}{\text{No of rotor pole teeth}} = \frac{360^\circ}{6} = 60^\circ$

So, one full step = Stator pole pitch- Rotor pole pitch

$$\text{Full step} = 45^\circ - 60^\circ = 15^\circ$$

Initial position of the rotor is shown in fig.1. With all switches S₁ to S₄ in OFF position. Now turn ON switch S₁, so that phase A is energized. This will make stator poles A and A' as north and south magnetic poles.

The stator poles attract rotor teeth and the rotor moves to occupy a position of minimum reluctance. Thus rotor teeth 1 and 4 take the rest positions as shown in figure.

Next, switch S_2 is turned ON and S_1 is turned OFF. Phase B excited, stator poles B and B' get magnetized. The rotor will move in such a direction to occupy the minimum reluctance path by travelling minimum angular distance. This is achieved by the movement of the rotor in counter clockwise direction by 15° is shown in fig

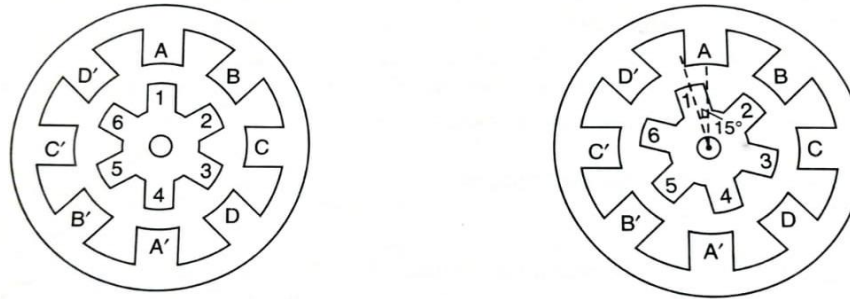


Fig. 2 Rotor position when phase A is excited & Rotor position when phase B is energized

With S_2 OFF and S_3 ON, the rotor moves again by 15° in the CCW direction and comes rest in position shown in Figure. S_4 ON and S_3 OFF, the rotor rotates by 15° in CCW direction and takes position shown in figure. By continuing this switching operation in the sequence $S_1, S_2, S_3, S_4, S_1 \dots$ rotor moves in CCW direction with step angle of 15°

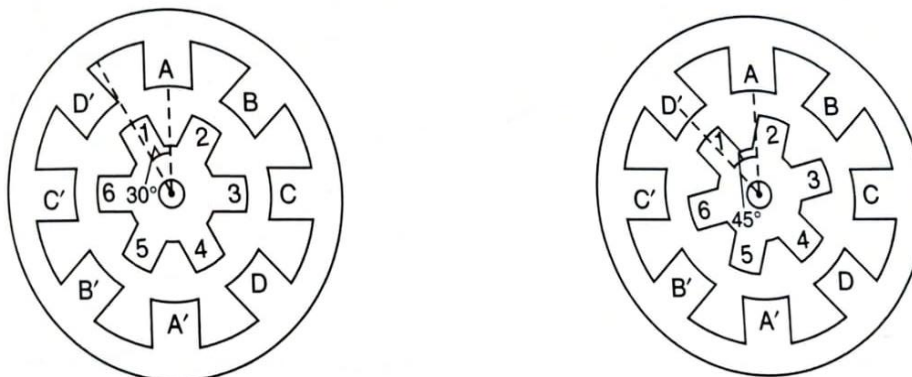


Fig. 3 Position of rotor after switching phase C & Rotor position after switching Phase D

The final position corresponds to the last switch in ON position. Finally, closed switch should remain in ON position to hold the rotor in that position and to prevent further movement due to external disturbance. The rotor remains in this position until and external torque greater than holding torque is applied on the motor

10.01.19

* Constructional details of $1-\phi$ I.M. There are two

important parts. They are

- 1) stator,
- 2) Rotor.

1) Stator: The stationary part in $1-\phi$ I.M is called stator. The stator has laminated construction made up of stampings. The stampings are slotted on its periphery to carry the winding called stator winding or main winding.

This is excited by a single- ϕ A.C supply.

The stampings are made up of silicon steel or iron. These stampings are laminated with Varnish or silica gel to reduce Eddy current losses.

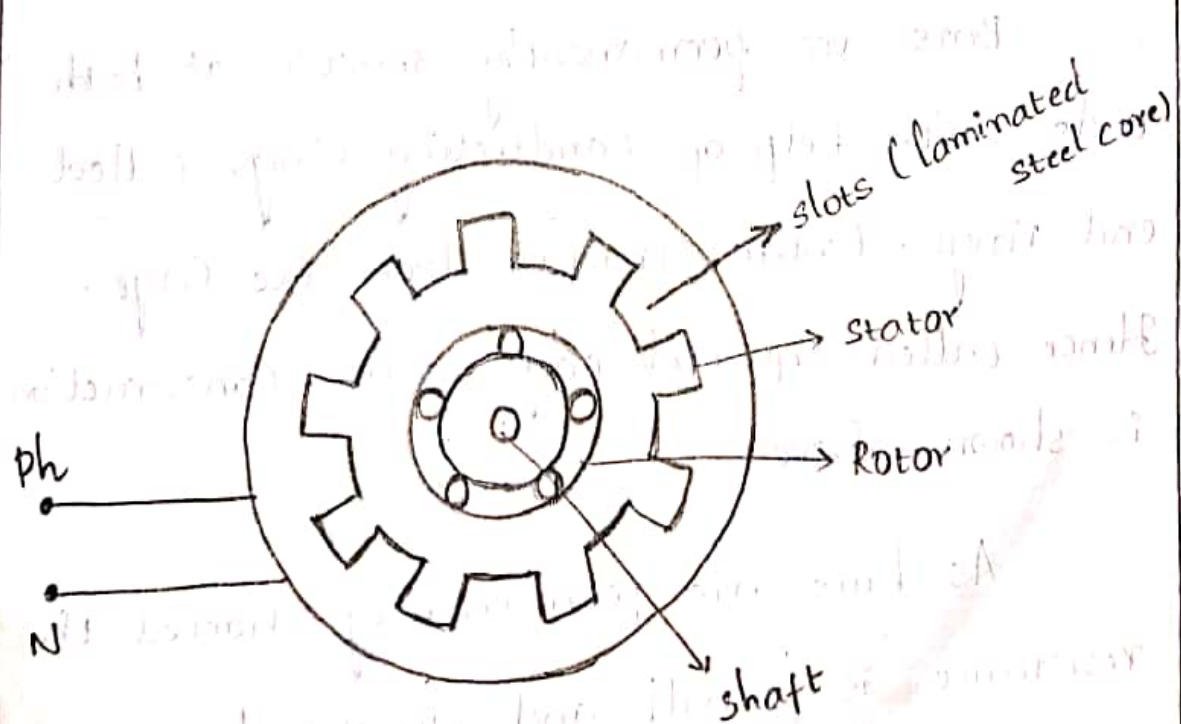
By using silicon steel stampings, minimizes the hysteresis losses. The stator winding has two main windings one is starting winding or auxiliary winding. Another one is Running winding or main winding.

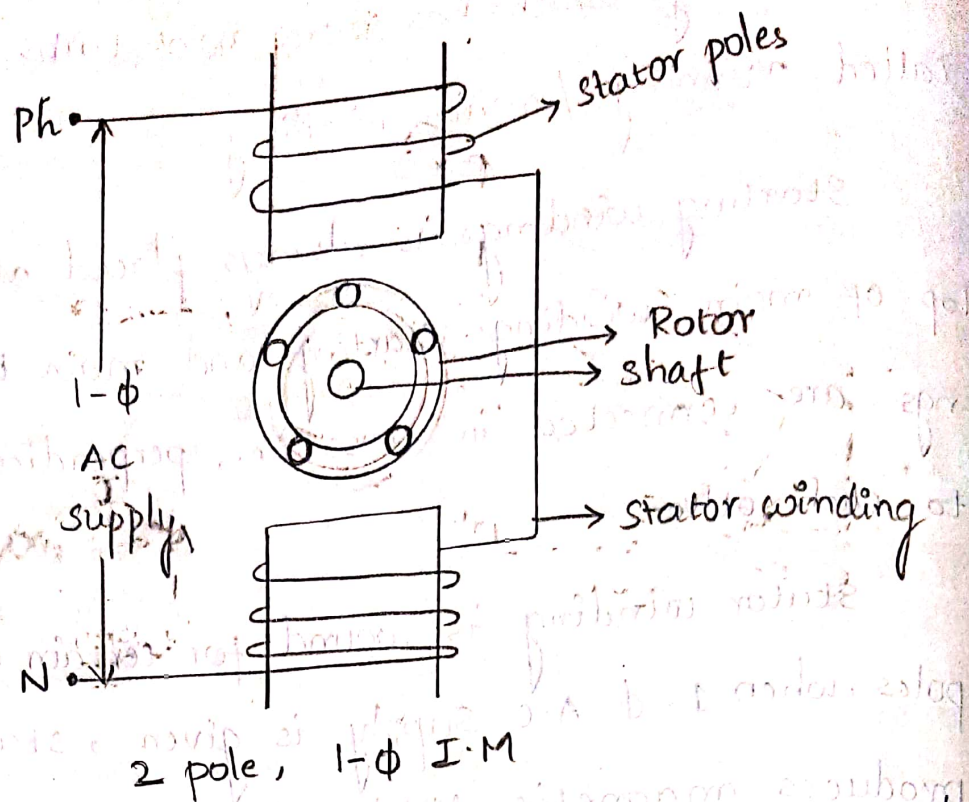
winding which has less no. of turns is called starting winding.

Winding which has more no. of turns is called running / main winding.

Starting winding is always placed at the top of main winding. Starting and main windings are connected in cross i.e., perpendicular to each other.

Stator winding is wound for certain no. of poles. When 1- ϕ A.C supply is given, stator produces magnetic field. The no. of poles for which stator winding is wound decides synchronous speed of motor i.e., $N_s = \frac{120f}{p}$, where 'f' is supply frequency, p is no. of poles.



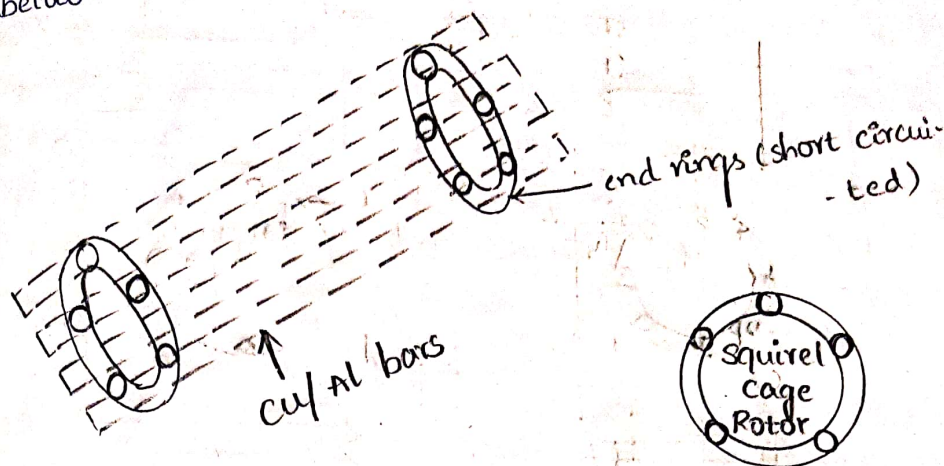


Rotor: Rotor is same as that of 3-φ Squirrel cage I.M. This type of rotor consists of uninsulated copper (or) Aluminium bars placed in the slots.

Bars are permanently shorted at both ends with help of conducting rings called end rings. Entire structure looks like Cage. Hence called squirrel cage rotor. Construction is shown above.

As bars are permanently shorted the resistance is small and air gap between stator and Rotor is made small.

Schematic representation of 1- ϕ I.M is shown below.

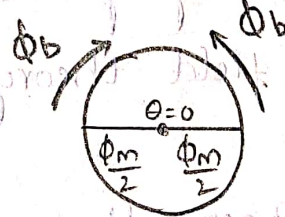
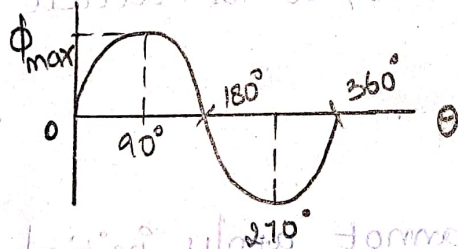


17/0/19

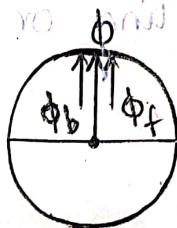
* Double Field Revolving / Ferrari theory /
two field / cross field theory:

1- ϕ AC supply $\rightarrow$ forward magnetic flux (ϕ_f) } same magnitude ($\phi_{max}/2$)
 $\rightarrow$ Backward magnetic flux (ϕ_b) } & rotating nature (opposite)

At starting, $\phi = 0$ ($\because \phi_f \sim \phi_b = 0$)



No force on stator at starting. so, there is no rotation. Hence, it is not self starting.



$$\phi_R = \frac{\phi_{max}}{2} + \frac{\phi_{max}}{2}$$

$$= \phi_{max}$$

3. Single Phase Induction Motor

There are three types of machines. They are

- 1) $1-\phi$ Asynchronous M/c, — $1-\phi$ I.M
- 2) $1-\phi$ synchronous M/c,
- 3) Commutation M/c.

Applications of $1-\phi$ I.M are;

1. Fans,
2. Blowers,
3. Compressors,
4. Hair Dryers,
5. Vacuum cleaners,
6. Washing Machines,
7. Pumping,
8. Lathes M/c's,
9. Drilling M/c's,
10. Diary M/c's,
11. Mixers,
12. Grinders,
13. Office tools / carpentary tools.